Compact Multilevel-Prior Tensor Representation for Hyperspectral Image Super-Resolution

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***Abstract*—Fusing a hyperspectral image (HSI) with a multispectral image acquired over the same scene, i.e., HSI super-resolution, has become a popular computational way to access the latent high-spatial–spectral-resolution image (HSSI). To date, a variety of fusion methods have been proposed, among which the tensor-based ones have testified that multiple priors, such as multidimensional low rankness and spatial total variation (TV) at multiple levels, effectively drive the fusion process. However, the existing tensor-based models can only effectively leverage one or two priors at one or two levels, since simultaneously incorporating multilevel priors inevitably increases model complexity. This introduces challenges in both balancing the weights of different priors and optimizing multiblock structures. Concerning this, we present a novel hyperspectral super-resolution (HSR) model compactly characterizing these multilevel priors of HSIs within the tensor framework. First, the proposed model decouples the spectral low rankness and spatial priors by casting the latent HSSI into spectral subspace and spatial maps via block term decomposition. Second, these spatial maps are stacked as the spatial tensor encoding the high-order spatial low rankness and smoothness priors, which are co-modeled via the proposed nonconvex mode-permuted tensor-correlated TV (NMP-t-CTV). Finally, we draw inspiration from the linearized alternating direction method of multipliers (LADMMs) to design an efficient algorithm to optimize the resulting model, theoretically proving its feasible-point convergence under mild conditions. Experiments on multiple datasets demonstrate the effectiveness of the proposed algorithm. The code implementation is available from https://github.com/WongYinJ**

***Index Terms*—Hyperspectral super-resolution (HSR), image fusion, low-rank tensor, nonconvex optimization, remote sensing, total variation (TV).**

I. Introduction

THE hyperspectral image (HSI) is much appreciated for its nanometer-level fine-grained spectral information which to a large extent facilitates many scene interpretation tasks, such as classification [1], [2], object tracking [3], and target detection [4], [5]. Unfortunately, acquiring high-spatial-resolution HSI at the hardware level remains a significant challenge due to the fundamental physical tradeoffs in the existing imaging systems. To be more specific, the desire for adequate photon capture to maintain photoelectric signal-to-noise-ratio necessitates larger detector units per spectral band in hyperspectral sensors, inherently reducing the spatial resolution of the acquired HSI. Such defect severely impairs the delineation of fine-scale features in the application scenarios above [6]. Consequently, fusing HSI with a corresponding multispectral image (MSI) that is easily captured at a much higher spatial resolution due to a much sparser spectral sampling rate [7], i.e., hyperspectral super-resolution (HSR), has become a popular software-level alternative to access to the latent high-spatial–spectral-resolution image (HSSI).

In this vein, numerous HSR models have been subsequently proposed. Song et al. [8] developed a dictionary-based fusion method using nonnegative matrix factorization to couple spectral and spatial information from HSI and MSI, respectively, achieving higher fidelity results than benchmarks on satellite data. A convex alternating direction method of multipliers (ADMMs)-based fusion method that leverages subspace modeling and edge-preserving vector total variation (TV) regularization to achieve HSR has been presented in [9], which jointly estimates the spatial and spectral responses that degrade the latent HSSI. Wei et al. [10] modeled HSR as the problem of solving a Sylvester equation maximizing the likelihood criterion. By leveraging the structural properties of the circulant and downsampling matrices inherent to the fusion problem, they derived a closed-form solution for the associated Sylvester equation, eliminating the need for iterative updates. Dian et al. [11] framed the fusion task as estimating spectral basis functions and their corresponding coefficients and integrated nonlocal spatial similarities, spectral unmixing priors, and a sparsity constraint into the fusion problem. Xue et al. [12] proposed a structured sparse low-rank representation model that effectively exploits spatial/spectral subspace relationships from higher level analysis, achieving superior HSR performance.

Despite their promising performance, all those methods fail to outrun the regime of matricizing the 3-D HSI data cube. Since matrix analysis relies on the conventional matrix product, which only captures row-column relationships, it struggles to model the intricate structures of high-dimensional data like HSI. Consequently, the tensor-based HSR approaches have gained increasing attention in HSI analysis [13]. Exploiting a coupled Tucker decomposition [14] paradigm, Li et al. [15] translated the spatial–spectral correlation of HSSI into the

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sparse prior on the core tensor and delivered a coupled sparse tensor factorization (CSTF) HSR model, which was later generalized for semi-blind HSR [16]. Chang et al. [17] proposed an HSI restoration model that simultaneously captures spectral–spatial nonlocal similarity and spectral correlation through third-order tensors, demonstrating how singular value reweighting significantly enhances tensor modeling capability and flexibility. Tensor convolution-like low-rank dictionary learning framework [18] enhanced high-dimensional image representation by explicitly exploiting the shift-invariant low-dimensional structures of tensor-valued data. As research progresses, it is acknowledged that HSI exhibits multidimensional structures that cannot be sufficiently characterized by imposing a single constraint on any individual factor. In light of this, Dian et al. [19] exploited the low-tensor-train (TT) [20] rank (LTTR) prior, formulated in a multirank form, of the grouped 4-D nonlocal patch tensors. They also proposed the generalized tensor nuclear norm (GTNN) [21] to promote flexibility when characterizing the multidimensional low-tensor tubal rank [22]. To avoid the computational burden inherited from patch clustering, Liu et al. [23] imposed the tensor trace norm [24] straightforwardly on the HSSI to cast the HSR task as a low Tucker rank tensor approximation (LRTA) problem. However, these methods formulate the multidimensional rank minimization problem as minimizing the nuclear norms of several matrices unfolded from the desired HSSI, incapable of explicitly analyzing the structures of the decomposed factors [25]. To overcome this limitation, Li et al. [26] proposed the anisotropic sparsity (AS) tensor norm to transform the multidimensional low rankness of the original tensor into the multidimensional sparsity of the Tucker core tensor. The resulting AS constrained low-rank tensor approximation (ASLA) framework achieves simultaneous multidimensional prior exploitation and factor structure pursuit under Tucker decomposition. Xu et al. [27] proposed the cascade-transform-based tensor nuclear norm (TNN) to alleviate the boundary effects and singleton domain limitation accompanied with the conventional TNN, which, noteworthily, was then generalized into a full-scale/multidimensional version, reducing its directional sensitivity. Another variant of TNN based on a nonlinear transform, along with its full-scale/multidimensional version, was presented in [28], breaking TNN's limitation in modeling complex data structures inherent in the linear transform. Additionally, TV [29], as an efficient tool modeling the spatial continuity equipped within diverse image modalities, has been successfully absorbed into tensor-based HSR frameworks, e.g., [26], [30], [31], [32], [33]. However, with multiprior being employed, these models inevitably increase by scale, which leads to the challenges of model selection and multiblock optimization. Thus, many of them suffer from hyperweight balancing and the loss of convergence guarantee. Besides, most of these methods only manage to extract those prior information from a single level, that is, from the original data or its factorized components. Although a multilevel characterization of sparse and low-rank priors was established [34], the intrinsic coupling between these two properties was not fully revealed, leaving room for further reducing the model scale. Hence, the comprehensive exploitation of multiprior at multilevel remains an open issue.

The advancement of computing devices has enabled the development of deep learning techniques. The HSR field has therefore witnessed a proliferation of data-driven methods. Nguyen et al. [35] proposed an unsupervised convolutional neural network (CNN)-based fusion framework that incorporates Stein's unbiased risk estimate (SURE) into a novel loss function, combining backprojection mean square error (mse) with SURE to approximate the projected mse between the fused and ground-truth HSSIs. Inspired by the smoothed particle hydrodynamics (SPHs) theory, Zhang et al. [36] analogized pixel motion to SPH particle dynamics, implementing multiscale smooth convolutions to preserve spectral information while using discretized Navier–Stokes equations to guide pixel motion for enhanced spatial edge clarity. Dian et al. [37] delivered a zero-shot learning (ZSL) HSR network that estimates sensor responses for realistic data simulation, also incorporating dimension reduction and model-based loss, thus overcoming data scarcity/generalization issues while maintaining high accuracy and efficiency. Ma et al. [38] replaced CNN with transformers to learn HSSI priors and unfolded a proximal gradient algorithm into a network, where self-attention captures global spatial interactions and 3D-CNN layers enhance spatial–spectral correlation modeling. Liu et al. [39] proposed a low-rank transformer for HSR to make both the inherent characteristics of HSSI and the learning ability of transformer into effect. Dian et al. [40] suggested an equivariant Bayesian variational inference framework for HSI computational imaging by integrating latent low-rank decomposition, equivariant imaging priors, and learnable physical degradation models to achieve accurate reconstruction. Wang et al. [41] introduced an unsupervised equivariant imaging framework with a learnable degradation function to reconstruct high-resolution HSIs without requiring ground truth (GT) or prior knowledge of the spectral response function. Nevertheless, the reliance on large-scale computational resources and high-quality training data remains a critical limitation hindering the practical application of data-driven methods.

In such context, this article presents an as compact as possible tensor model for the HSR task, to concisely realize multilevel-prior representation, which further facilitates a convergence-guaranteed iterative algorithm. Specifically, the contributions are three-folds.

1) We present the nonconvex mode-permuted tensor-correlated TV (NMP-t-CTV), which builds upon the t-CTV framework [42] by introducing a mode-permuted strategy and extending it from a convex approximation to a nonconvex formulation. By imposing it on the stacked spatial tensor acquired from the block-term decomposition (BTD) [43], we theoretically justify that the resulting model can cross-represent multilevel-priors equipped within the latent HSSI in a compact form.
2) Inspired by the linearized ADMMs (LADMMs) [44], [45], we customize an optimization framework for the proposed model that successfully deals with the high dimensionality of the spatial tensor. With a mild

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assumption that some multipliers are bounded, we manage to theoretically guarantee the convergence of the optimization framework to the feasible point, which also benefits from the compactness of the proposed model.

3) A broad experimental analysis demonstrates the practical efficacy of the proposed compact multilevel-prior tensor representation (CMlpTR).

The remainder of this article is organized as follows. Section II familiarizes the readers with the notation and the foundational concepts related to this article. Section III presents the proposed CMlpTR and the derivation of the optimization framework. Section IV shows the experimental results. Finally, Section V draws the conclusions.

II. Preparation

A. Notations

Throughout this article, we denote a scalar as a, a vector as $\boldsymbol{a}$, a matrix as $\boldsymbol{A}$, and a tensor as $\mathcal{A}$. $\|\boldsymbol{A}\|_F, \|\boldsymbol{A}\|, \|\boldsymbol{A}\|_1$, and $\|\boldsymbol{A}\|_*$ denote the Frobenius, the l_1, and the nuclear norms of a matrix, respectively. Particularly, we use $\boldsymbol{I}_N$ to denote the $N \times N$ identity matrix, $\boldsymbol{F}_N$ to denote the $N \times N$ discrete Fourier transformation matrix, and $\boldsymbol{D}_N$ to denote the $(N-1) \times N$ first-order difference matrix, i.e.,

$$\boldsymbol{D}_N = \begin{bmatrix} 1 & -1 & & \\ & 1 & -1 & \\ & & \ddots & \ddots \\ & & & 1 \quad -1 \end{bmatrix}.$$

Let $\boldsymbol{A}^*$ be the conjugate transpose of $\boldsymbol{A}$. Then, we can denote the anisotropic TV (ATV) norm of a matrix $\boldsymbol{A} \in \mathbb{R}^{M \times N}$ as

$$\|\boldsymbol{A}\|_{ATV} \triangleq \|\boldsymbol{D}_M \boldsymbol{A}\|_1 + \left\|\boldsymbol{A}\boldsymbol{D}_N^*\right\|_1.$$

For a three-way tensor $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, we use $\mathcal{A}[i_1, i_2, i_3]$ to denote the (i_1, i_2, i_3)th element of $\mathcal{A}$ and $\mathcal{A}_{:,:,i_3}$ to denote its i_3th frontal slice. Its Frobenius norm is then denoted by $\|\mathcal{A}\|_F$ and defined as $\|\mathcal{A}\|_F = \left(\sum_{i_1,i_2,i_3} \mathcal{A}^2[i_1, i_2, i_3]\right)^{1/2}$. Denoting the mode-$n$ product [14] with $\times_n$, the TV and ATV norms of $\mathcal{A}$ are defined as

$$\|\mathcal{A}\|_{TV} \triangleq \sqrt{\|\mathcal{A} \times_1 \boldsymbol{D}_M\|_F^2 + \|\mathcal{A} \times_2 \boldsymbol{D}_N\|_F^2}$$
$$\|\mathcal{A}\|_{ATV} \triangleq \|\mathcal{A} \times_1 \boldsymbol{D}_M\|_1 + \|\mathcal{A} \times_2 \boldsymbol{D}_N\|_1.$$

By $\circ$, we denote the outer product [43, Definition 1.3]. Besides, with a little notation abuse, we also denote the tensor conjugate transpose [46] of $\mathcal{A}$ with $\mathcal{A}^*$. To facilitate presentation, we use $\texttt{svds}(\cdot)$ and $\texttt{permute}(\cdot)$ to denote two operators with the same functionality of their MATLAB namesakes.

B. Tensor Preliminaries

The tools to depict the tensor rank are various, among which this article employs the tensor singular value decomposition (t-SVD), whose definition is as follows.

Definition 1 (t-SVD [47]): For a three-way tensor $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, there exist $\mathcal{U} \in \mathbb{R}^{I_1 \times I_1 \times I_3}$, $\mathcal{S} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, $\mathcal{V} \in \mathbb{R}^{I_2 \times I_2 \times I_3}$, such that

$$\mathcal{A} = \mathcal{U} \star \mathcal{S} \star \mathcal{V}^*$$

in which $\star$ denotes the t-product [48], $\mathcal{U}$ and $\mathcal{V}$ are the orthogonal tensors [46], and $\mathcal{S}$ is an f-diagonal tensor [46].

Accordingly, denoting the Fourier transformation of $\mathcal{S}$ along mode-3 as $\overline{\mathcal{S}}$, i.e., $\overline{\mathcal{S}} = \mathcal{S} \times_3 \boldsymbol{F}_L$, then the TNN of $\mathcal{A}$ is defined as follows.

Definition 2 (TNN [46]): For the three-way tensor $\mathcal{A}$ in Definition 1, its TNN can be written as

$$\|\mathcal{A}\|_* \triangleq \frac{1}{I_3} \sum_{j=1}^{I_3} \sum_{i=1}^{\min\{I_1, I_2\}} \overline{\mathcal{S}}[i, i, j].$$

Hence, we can define t-CTV as follows.

Definition 3 (t-CTV [42]): For a three-way tensor $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, its t-CTV can be written as

$$\|\mathcal{A}\|_{\circlearrowright *} \triangleq \frac{1}{|\mathfrak{C}|} \sum_{n \in \mathfrak{C} \subset \{1,2,3\}} \left\|\mathcal{A} \times_n \boldsymbol{D}_{I_n}\right\|_*$$

in which $|\mathfrak{C}|$ denotes the cardinality of $\mathfrak{C}$ and $\mathcal{A} \times_n \boldsymbol{D}_{I_n}$ is called the mode-n gradient tensor.

To improve the approximation tightness of TNN to the t-SVD rank [42], several works have proposed to use nonconvex surrogates; thus, the nonconvex tensor pseudo nuclear norm (NTPNN) is defined as follows.

Definition 4 (NTPNN [49]): For the three-way tensor $\mathcal{A}$ in Definition 1, its NTPNN is

$$\|\mathcal{A}\|_{*,\psi} \triangleq \frac{1}{I_3} \sum_{j=1}^{I_3} \sum_{i=1}^{\min\{I_1, I_2\}} \psi\left(\overline{\mathcal{S}}[i, i, j]\right)$$

in which $\psi(\cdot) : [0, +\infty) \to [0, +\infty)$ is some nonconvex surrogate.

Note that both TNN and NTPNN are restricted to the third mode, which lacks flexibility dealing with the multidimensional structure of tensors. This article then exploits the idea of GTNN to comprehensively consider the all-around low-t-SVD-rankness (LTSVDR) along all modes, for which we define the NTPNN for mode-1 and mode-2.

Definition 5 (Mode-$\{1, 2\}$ NTPNN): For a three-way tensor $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, we denote a permutation operator $\mathtt{P}_n(A) \triangleq \texttt{permute}(\mathcal{A}, [3-n, 3, n])$. Then, the mode-$n$ NTPNN of $\mathcal{A}$ is

$$\|\mathcal{A}\|_{*,\psi}^{n} \triangleq \|\mathtt{P}_n(A)\|_{*,\psi}, n = 1, 2.$$

C. HSR Problem Formulation

For the convenience of derivation, we model the degradation from HSSI to HSI and MSI following TF-HSR [50]. For the sake of consistency, this is reported here.

Definition 6 (TF-HSR [50]): Given the HSI $\mathcal{X} \in \mathbb{R}^{i_1 \times i_2 \times I_3}$, the MSI $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times i_3}$, the spatial-degradation matrices $\boldsymbol{P}_1 \in \mathbb{R}^{i_1 \times I_1}$ and $\boldsymbol{P}_2 \in \mathbb{R}^{i_2 \times I_2}$, and the spectral-degradation matrix $\boldsymbol{P}_3 \in \mathbb{R}^{i_3 \times I_3}$ with $i_1 < I_1$, $i_2 < I_2$, and $i_3 < I_3$, the TF-HSR problem estimates the most appropriate HSSI $\mathcal{Z} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, such that

$$\begin{aligned} \mathcal{X} &= \mathcal{Z} \times_1 \boldsymbol{P}_1 \times_2 \boldsymbol{P}_2 \\ \mathcal{Y} &= \mathcal{Z} \times_3 \boldsymbol{P}_3. \end{aligned} \tag{1}$$

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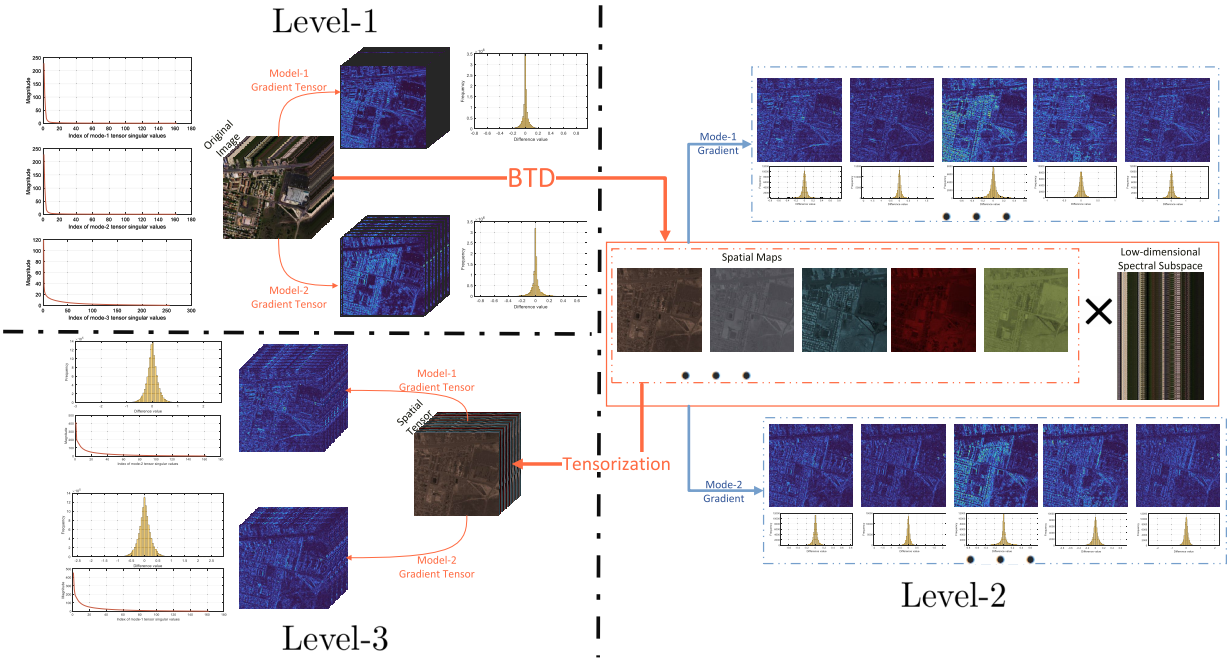


Fig. 1. Multilevel-prior observation. Level-1: original data domain; level-2: factorization domain; level-3: enhanced tensorization domain.

III. Methodology

A. Multilevel-Prior Observation of HSSI

Since a real HSSI $\mathcal{Z}$ is by no means accessible, we stay in line with other HSR researches (e.g., [17], [32], [50]) to observe the priors of HSI as a surrogate. First, as shown by the Level-1 part of Fig. 1, in the original HSI, there underlies the multimode LTSVDR, i.e., the tensor singular values of $\mathtt{P}_1(\mathcal{Z})$, $\mathtt{P}_2(\mathcal{Z})$, and $\mathcal{Z}$ exhibit rapid decay. Besides, taking into account the spatial TV prior, the mode-1,2 gradient tensors of $\mathcal{Z}$ should exhibit sparsity.

Furthermore, to develop deeper-level priors for HSIs within their factorization domain, we turn to the BTD framework. Among the many tensor decomposition approaches available, the BTD is particularly well-suited to our objectives. It aligns naturally with the physical formation model of HSI, where each pixel is composed of endmember signatures and their corresponding abundance maps [51]. This intrinsic connection enhances the potential for extracting physically meaningful priors. Moreover, the BTD is supported by a solid theoretical foundation [43], which strengthens the pursuit of a theoretically robust modeling framework. As thus, following the BTD paradigm, the HSI can be decomposed into a spectral subspace and a series of spatial maps, which is illustrated by the Level-2 part of Fig. 1 and written as:

$$\mathcal{Z} = \mathcal{A} \times_3 \boldsymbol{S} = \sum_{r=1}^{R} \mathcal{A}_{:,:,r} \circ \boldsymbol{s}_r \tag{2}$$

where $\mathcal{A} \in \mathbb{R}^{I_1 \times I_2 \times R}$ is the spatial tensor whose frontal slices $\{\mathcal{A}_{:,:,r}\}_{r=1}^{R}$ are the spatial maps and $\boldsymbol{S} \in \mathbb{R}^{I_3 \times R}$ is the spectral subspace with $\boldsymbol{s}_r$ its rth column. Thereupon, the spectral low rankness of the HSSI is reflected by the low dimensionality of $\boldsymbol{S}$, while the spatial smoothness is encoded in the horizontal and vertical gradients of the spatial maps.

Moreover, Wang et al. [52] have suggested that by inspecting the stacked spatial tensor $\mathcal{A}$ instead of the spatial maps, we get the enhanced version of spatial priors. To this end, and also inspired by t-CTV, we examine the spatial TV and low rankness both from the gradient tensors of $\mathcal{A}$, as visualized by the Level-3 part of Fig. 1.

Thus, taking $\mathcal{Z} = \mathcal{A} \times_3 \boldsymbol{S}$ into (1), the HSR problem is reformulated as

$$\text{Find } \mathcal{A}, \boldsymbol{S}, \quad \text{s.t., } \mathcal{X} = \mathcal{A} \times_1 \boldsymbol{P}_1 \times_2 \boldsymbol{P}_2 \times_3 \boldsymbol{S} \quad \mathcal{Y} = \mathcal{A} \times_3 \boldsymbol{P}_3 \boldsymbol{S}.$$

B. Multilevel-Prior Tensor Representation

As it is observed that multilevel priors are equipped within the HSSI in the BTD paradigm, we now establish a constraint to compactly represent these multilevel priors. While t-CTV jointly enforces low rankness and smoothness via a nuclear norm constraint on gradient tensors, it computes the norm uniformly across all modes. This fixed-mode pattern lacks the flexibility to capture the distinct structural priors along different dimensions of HSSI data. To address this limitation, we propose NMP-t-CTV, which adaptively imposes mode-specific low-rank constraints on gradient tensors for more accurate prior characterization.

Definition 7 (NMP-t-CTV): For the spatial tensor $\mathcal{A}$ defined in (2), its NMP-t-CTV, denoted as $\|\mathcal{A}\|_{\widetilde{*},\psi}$, is defined as

$$\|\mathcal{A}\|_{\widetilde{*},\psi} = \frac{1}{2} \sum_{n=1}^{2} \left\| \mathcal{A} \times_n \boldsymbol{D}_{I_n} \right\|_{\overset{3-n}{*},\psi}.$$

In the definition above, the mode along which the LTSVDR is employed with respect to each gradient tensor is designed. Specifically, for the mode-1 gradient tensor, its mode-2 LTSVDR is considered, while the mode-1 LTSVDR is imposed on the mode-2 gradient tensor. This facilitates the

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cross-representation of multilevel priors in a compact way. Besides, we employ the NTPNN in replacement of the convex TNN to achieve tighter approximation of the tensor rank. By doing so, the proposed CMlpTR model for HSR is as follows:

$$\begin{aligned}\min_{\mathcal{A},\boldsymbol{S}}\ & \frac{1}{2}\sum_{n=1}^{2}\left\|\mathcal{A}\times_n \boldsymbol{D}_{I_n}\right\|_{\substack{3-n\\ *,\psi}}\\ \text{s.t.},\ & \mathcal{X}=\mathcal{A}\times_1 \boldsymbol{P}_1\times_2 \boldsymbol{P}_2\times_3 \boldsymbol{S}\\ & \mathcal{Y}=\mathcal{A}\times_3 \boldsymbol{P}_3\boldsymbol{S}.\end{aligned}\tag{3}$$

Model (3) is then designed to compactly co-represent the aforementioned multilevel priors with only two-block variables and two constraints. To help justifying its ability, we deliver Proposition 1.

Proposition 1: Regarding (2), if $\boldsymbol{S}\in\mathbb{R}^{I_3\times R}$ is semi-unitary, i.e., $\boldsymbol{S}^*\boldsymbol{S}=\boldsymbol{I}_R$, there follows

$$\begin{aligned}&rank_{\substack{3-n\\ t}}(\mathcal{Z})-1\le rank_{\substack{3-n\\ t}}\left(\mathcal{A}\times_n \boldsymbol{D}_{I_n}\right)\le rank_{\substack{3-n\\ t}}(\mathcal{Z})\\ &\qquad n=1,2.\end{aligned}\tag{4}$$

If we further have that $\mathcal{A}$ is bounded and $\psi(\cdot)$ is endowed with the properties that:
Suppose further that $(\mathcal{A})$ is bounded and that $(\psi(\cdot))$ satisfies the following properties:

P1: $\psi(\cdot)$ is concave and nondecreasing, with $\psi(0)=0$.

P2: $\psi'(\cdot)$ is convex and nonincreasing, with $\lim_{x\to 0^+}\psi'(x)<\infty$; then, the NMP-t-CTV is compatible with the TV norms, i.e., $\exists\, a_1,a_2,a_3,a_4>0$, such that

$$\begin{aligned}a_1\|\mathcal{A}\|_{TV}\le\|\mathcal{A}\|_{*,\psi}^{\sim}\le a_2\|\mathcal{A}\|_{TV}\\ a_3\|\mathcal{A}\|_{ATV}\le\|\mathcal{A}\|_{*,\psi}^{\sim}\le a_4\|\mathcal{A}\|_{ATV}.\end{aligned}\tag{5}$$

Proof: Deferred to Section I in Supplemental Material. ■

P1 and **P2** can be traced back to studies on nonconvex surrogates for rank minimization [52], [53]. **P1** is essential for establishing the equivalence inequalities in (5). This allows the proposed NMP-t-CTV to simultaneously encode both low rankness and spatial smoothness—a key contribution of our multilevel prior representation. **P2** guarantees that the proximal mapping of $\psi(\cdot)$ is well-defined and efficiently solvable via generalized accelerated iterative shrinkage [53], which underpins the update steps for (14).

As a result, the co-representation ability of our CMlpTR can be explained as follows.

1) Since $\boldsymbol{S}$ encodes the spectral subspace of HSI/HSSI, the spectral low rankness of $\mathcal{Z}$ can be modeled by controlling the dimensionality of $\boldsymbol{S}$.
2) According to [42, Remark 1], the multidimensional mode-{1,2} LTR of $\mathcal{A}$ is well captured.
3) Based on (4), our CMlpTR can express the spatial low rankness of the original $\mathcal{Z}$, which together with 1) realizes the co-representation of the multidimensional LTR within $\mathcal{Z}$.
4) Equation (5) asserts that the spatial TV prior of HSI/HSSI, reflected on the spatial tensor $\mathcal{A}$, is well characterized. Moreover, since we have $\|\mathcal{A}\|_{ATV}=\sum_{r=1}^{R}\|\mathcal{A}_{:,:,r}\|_{ATV}$, the TV prior on the 2-D spatial maps of the BTD level is also expressed.

Besides, it is known from Proposition 1 that to take 1)–4) into effect, some additional conditions should be satisfied by $\boldsymbol{S}$ and $\psi(\cdot)$. Thus, we specify the nonconvex surrogate as $\psi(x)=(\log(\gamma x+1))/(\log(\gamma+1))$ [54] and add the semi-unitary constraint on $\boldsymbol{S}$ to obtain

$$\begin{aligned}\min_{\mathcal{A},\boldsymbol{S}}\ & \frac{1}{2}\sum_{n=1}^{2}\left\|\mathcal{A}\times_n \boldsymbol{D}_{I_n}\right\|_{\substack{3-n\\ *,\psi}}\\ \text{s.t.},\ & \mathcal{X}=\mathcal{A}\times_1 \boldsymbol{P}_1\times_2 \boldsymbol{P}_2\times_3 \boldsymbol{S}\\ & \mathcal{Y}=\mathcal{A}\times_3 \boldsymbol{P}_3\boldsymbol{S}\\ & \boldsymbol{S}^*\boldsymbol{S}=\boldsymbol{I}_R.\end{aligned}\tag{6}$$

C. Optimization Algorithm

To find a solution to Problem (6), it is noticed that auxiliary variables are unavoidable in solving the $\mathcal{A}$-subproblems, leading to the expansion of the problem scale. Wang et al. [50] demonstrate that the subspace and coefficient can be determined in a sequential manner without hurting the model performance. Thus, we extract the semi-unitary spectral subspace $\boldsymbol{S}$ simply from HSI $\mathcal{X}$ as

$$[\boldsymbol{S},\sim,\sim]=\mathsf{svds}\left(\mathcal{X}_{[3]},R\right).$$

Then, the problem can be reduced as

$$\begin{aligned}\min_{\mathcal{A}}\ & \frac{1}{2}\sum_{n=1}^{2}\left\|\mathcal{A}\times_n \boldsymbol{D}_{I_n}\right\|_{\substack{3-n\\ *,\psi}}\\ \text{s.t.},\ & \mathcal{X}=\mathcal{A}\times_1 \boldsymbol{P}_1\times_2 \boldsymbol{P}_2\times_3 \boldsymbol{S}\\ & \mathcal{Y}=\mathcal{A}\times_3 \boldsymbol{P}_3\boldsymbol{S}.\end{aligned}$$

To solve it, we introduce some auxiliary variables, i.e., $\mathcal{G}_1$ and $\mathcal{G}_2$, and we obtain

$$\begin{aligned}\min_{\mathcal{A},\{\mathcal{G}_n\}_{n=1,2}}\ & \frac{1}{2}\sum_{n=1}^{2}\|\mathcal{G}_n\|_{\substack{3-n\\ *\ \psi}}\\ \text{s.t.},\ & \mathcal{X}=\mathcal{A}\times_1 \boldsymbol{P}_1\times_2 \boldsymbol{P}_2\times_3 \boldsymbol{S}\\ & \mathcal{Y}=\mathcal{A}\times_3 \boldsymbol{P}_3\boldsymbol{S}\\ & \mathcal{G}_n=\mathcal{A}\times_n \boldsymbol{D}_{I_n}, n=1,2.\end{aligned}\tag{7}$$

Usually, problem like (7) is easily solved by ADMM [55]. However, the $\mathcal{A}$-subproblem suffers from the high dimensionality of $\mathcal{A}$, compelling us to make adjustment to its ADMM-induced updating steps. The resulting algorithm can be viewed as a composition of LADMM. In the following, we elaborate on it and provide its convergence analysis in Theorem 1.

Let us start defining the augmented Lagrangian function of (7) as

$$\begin{aligned}&L\left(\mathcal{A},\ \{\mathcal{G}_n\}_{n=1,2},\ \mathcal{M}_x,\ \mathcal{M}_y,\ \{\mathcal{M}_n\}_{n=1,2}\right)\\ &=\frac{1}{2}\sum_{n=1}^{2}\|\mathcal{G}_n\|_{\substack{3-n\\ *,\psi}}+\frac{\rho}{2}\Bigg(\left\|\mathcal{X}+\frac{\mathcal{M}_x}{\rho}-\mathcal{A}\times_1 \boldsymbol{P}_1\right.\\ &\quad\left.\times_2 \boldsymbol{P}_2\times_3 \boldsymbol{S}\right\|_F^2+\left\|\mathcal{Y}+\frac{\mathcal{M}_y}{\rho}-\mathcal{A}\times_3 \boldsymbol{P}_3\boldsymbol{S}\right\|_F^2\\ &\quad+\sum_{n=1}^{2}\left\|\mathcal{G}_n+\frac{\mathcal{M}_n}{\rho}-\mathcal{A}\times_n \boldsymbol{D}_{I_n}\right\|_F^2\Bigg)\end{aligned}$$

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$$-\frac{1}{2\rho}\left(\left\|\mathcal{M}_x\right\|_F^2+\left\|\mathcal{M}_y\right\|_F^2+\sum_{n=1}^{2}\left\|\mathcal{M}_n\right\|_F^2\right) \tag{8}$$

where $\{\mathcal{M}_x, \mathcal{M}_y, \{\mathcal{M}_n\}_{n=1,2}\}$ are the Lagrangian multipliers and ρ is the penalty parameter.

1) *$\mathcal{A}$-Subproblem:* Fixing $\{\mathcal{G}_n\}_{n=1,2}$, $\mathcal{A}$ is updated via solving

$$\begin{aligned}\min_{\mathcal{A}} L_1(\mathcal{A}) \triangleq &\left\|\mathcal{X}+\frac{\mathcal{M}_x}{\rho}-\mathcal{A}\times_1\boldsymbol{P}_1\times_2\boldsymbol{P}_2\times_3\boldsymbol{S}\right\|_F^2\\ &+\left\|\mathcal{Y}+\frac{\mathcal{M}_y}{\rho}-\mathcal{A}\times_3\boldsymbol{P}_3\boldsymbol{S}\right\|_F^2\\ &+\sum_{n=1}^{2}\left\|\mathcal{G}_n+\frac{\mathcal{M}_n}{\rho}-\mathcal{A}\times_n\boldsymbol{D}_{I_n}\right\|_F^2.\end{aligned} \tag{9}$$

Typically, (9) can be easily solved by setting its gradient to zero. However, the high dimensionality of $\mathcal{A}$ is a curse for doing so. Thus, inspired by LADMM, we approximate its solution by a simple one-step gradient descend, i.e.,

$$\mathcal{A}\leftarrow\mathcal{A}-\frac{\nabla L_1(\mathcal{A})}{\tau} \tag{10}$$

in which the gradient is calculated as

$$\begin{aligned}&\nabla L_1(\mathcal{A})\\ &=2\Bigg(\mathcal{A}\times_1\boldsymbol{P}_1^*\boldsymbol{P}_1\times_2\boldsymbol{P}_2^*\boldsymbol{P}_2+\mathcal{A}\times_3\boldsymbol{S}^*\boldsymbol{P}_3^*\boldsymbol{P}_3\boldsymbol{S}\\ &\quad+\sum_{n=1}^{2}\mathcal{A}\times_n\boldsymbol{D}_{I_n}^*\boldsymbol{D}_{I_n}-\left(\mathcal{X}+\frac{\mathcal{M}_x}{\rho}\right)\times_1\boldsymbol{P}_1^*\\ &\quad\times_2\boldsymbol{P}_2^*\times_3\boldsymbol{S}^*-\left(\mathcal{Y}+\frac{\mathcal{M}_y}{\rho}\right)\times_3\boldsymbol{S}^*\boldsymbol{P}_3^*-\\ &\quad\sum_{n=1}^{2}\left(\mathcal{G}_n+\frac{\mathcal{M}_n}{\rho}\right)\times_n\boldsymbol{D}_{I_n}^*\Bigg)\end{aligned} \tag{11}$$

and the updating stride τ should satisfy the Lipschitz continuity condition that

$$\left\|\nabla L_1(\mathcal{A})-\nabla L_1(\mathcal{A}')\right\|_F\le\tau\left\|\mathcal{A}-\mathcal{A}'\right\|_F \tag{12}$$

for any $\mathcal{A}, \mathcal{A}'$. To this end, we set

$$\tau=2\left(\|\boldsymbol{P}_1\|^2\|\boldsymbol{P}_2\|^2+\|\boldsymbol{P}_3\boldsymbol{S}\|^2+\|\boldsymbol{P}_1\|^2+\|\boldsymbol{P}_2\|^2\right). \tag{13}$$

Note that a single application of (10) does not yield the optimal solution to subproblem (9). Nevertheless, by incorporating it into the overall iterative procedure, we conjecture that the resulting sequence converges to a satisfactory solution of the original problem (7). This conjecture is subsequently verified by Theorem 1.

2) *$\mathcal{G}_n$-Subproblem, $n=1,2$:* Fixing $\mathcal{A}$, $\mathcal{G}_n$ is updated by solving

$$\begin{aligned}\min_{\mathcal{G}_n} L_{n+1}(\mathcal{G}_n) \triangleq &\|\mathcal{G}_n\|_{*,\psi}^{\mathbb{P}_{3-n}}\\ &+\rho\left\|\mathcal{G}_n+\frac{\mathcal{M}_n}{\rho}-\mathcal{A}\times_n\boldsymbol{D}_{I_n}\right\|_F^2\end{aligned} \tag{14}$$

which is equivalent to

$$\begin{aligned}\min_{\mathcal{G}_n} L_{n+1}(\mathcal{G}_n) \triangleq &\|\mathbb{P}_{3-n}(\mathcal{G}_n)\|_{*,\psi}^{n}\\ &+\rho\left\|\mathbb{P}_{3-n}(\mathcal{G}_n)+\mathbb{P}_{3-n}\left(\frac{\mathcal{M}_n}{\rho}-\mathcal{A}\times_n\boldsymbol{D}_{I_n}\right)\right\|_F^2.\end{aligned}$$

By [52, Th. 2], we first obtain the t-SVD of $\mathbb{P}_{3-n}(\mathcal{A}\times_n\boldsymbol{D}_{I_n}-\mathcal{M}_n/\rho)$ as $\mathcal{U}\star\mathcal{S}\star\mathcal{V}^*$; then, the solution to (14) is $\mathbb{P}_{3-n}^{-1}(\mathcal{U}\star\mathcal{T}\star\mathcal{V}^*)$, where τ is an f-diagonal tensor with

$$\begin{aligned}\overline{\mathcal{T}}[j,j,i]=\arg\min_{x\ge0}\psi(x)+\rho\left(x-\overline{\mathcal{S}}[j,j,i]\right)^2\\ i=1,2,\ldots,I_{3-n},\ j=1,2,\ldots,\min\{I_n,R\}\end{aligned}$$

which is solved via the generalized accelerated iterative shrinkage [53], and the remaining elements being zero.

3) *Auxiliary Variables:* Following the convention of the ADMM framework, we update the multipliers as:

$$\begin{aligned}&\mathcal{M}_x\leftarrow\mathcal{M}_x+\rho(\mathcal{X}-\mathcal{A}\times_1\boldsymbol{P}_1\times_2\boldsymbol{P}_2\times_3\boldsymbol{S})\\ &\mathcal{M}_y\leftarrow\mathcal{M}_y+\rho(\mathcal{Y}-\mathcal{A}\times_3\boldsymbol{P}_3\boldsymbol{S})\\ &\mathcal{M}_n\leftarrow\mathcal{M}_n+\rho\left(\mathcal{G}_n-\mathcal{A}\times_n\boldsymbol{D}_{I_n}\right),\ n=1,2.\end{aligned} \tag{15}$$

To help forcing convergence, we update the penalty parameter as $\rho\leftarrow\nu\rho$, where $\nu>1$.

D. Complexity and Convergence Analysis

The overall optimization algorithm is summarized in Algorithm 1.

In the updating of the spatial tensor $\mathcal{A}$, the main complexity lies in the calculation of the gradient, i.e., (11), whose complexity is

$$\begin{aligned}C_1\triangleq\mathcal{O}\Big(&2*\left(I_1^2I_2R+I_1I_2^2R\right)+R^2I_1I_2+i_1i_2I_1I_3\\ &+i_2I_1I_2I_3+I_1I_2I_3R+i_3I_1I_2R+(I_1-1)I_1I_2R\\ &+I_1(I_2-1)I_2R\Big).\end{aligned}$$

We further note that, if (11) is solved by zeroing the gradient, the corresponding complexity would be $\mathcal{O}((I_1I_2R)^3)$, which is clearly infeasible for general devices.

To update the auxiliary variables $\mathcal{G}_1$ and $\mathcal{G}_2$, their t-SVDs drive the complexity, which in total is

$$C_2\triangleq\mathcal{O}\left((I_1(I_2-1)+I_2(I_1-1))R^2\right).$$

The last part of the complexity comes from the tensor-matrix product in the updating of multipliers, i.e., (15)

$$C_3\triangleq\mathcal{O}(I_1I_2R(i_1+i_2+i_3+I_1-1+I_2-1+I_3)).$$

Therefore, the overall complexity of Algorithm 1 is $C_1+C_2+C_3$.

Algorithm 1 does not fall into any category of an existing convergence-guaranteed optimization paradigm. Its rough structure resembles the ADMM framework while the solution to some subproblems is customized following the LADMM. Besides, the employment of nonconvex surrogate brings more challenges to its convergence issue. To tackle this, we provide the convergence analysis as follows.

Theorem 1: Denoting the sequence generated by Algorithm 1 as

$$\mathfrak{S}_t=\left\{\mathcal{A}_t,\{\mathcal{G}_{n,t}\}_{n=1,2},\mathcal{M}_{x,t},\mathcal{M}_{y,t},\{\mathcal{M}_{n,t}\}_{n=1,2}\right\}_{t\in\mathbb{N}}$$

given that $\{\mathcal{M}_{x,t},\mathcal{M}_{y,t}\}_{t\in\mathbb{N}}$ are bounded, it follows that: Let $[\mathcal{S}_t=\left\{\mathcal{A}_t,\{\mathcal{G}_{n,t}\}_{n=1,2},\mathcal{M}_{x,t},\mathcal{M}_{y,t},\{\mathcal{M}_{n,t}\}_{n=1,2}\right\}_{t\in\mathbb{N}}]$ denote

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Algorithm 1 CMlpTR Optimization Algorithm

Input: The observed HSI $\mathcal{X}$, the MSI $\mathcal{Y}$, the spatial-degradation matrices $\boldsymbol{P}_1$ and $\boldsymbol{P}_2$, the spectral-degradation matrix $\boldsymbol{P}_3$, the subspace dimensionality R, the surrogate parameter γ.

Output: Estimated HSSI $\widehat{\mathcal{Z}}$.

Initialize penalty parameters $\rho > 0$, $\nu > 1$, the convergence tolerance $\epsilon > 0$, the difference matrices $\boldsymbol{D}_{I_1}$ and $\boldsymbol{D}_{I_2}$.
Set $\mathcal{A}, \mathcal{G}_1, \mathcal{G}_2, \mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_x$ and $\mathcal{M}_y$ to $\mathbf{0}$.
$[\boldsymbol{S}, \sim, \sim] = \mathsf{svds}(\mathcal{X}_{[3]}, R)$.
while $\max\left\{\left\|\mathcal{X} - \mathcal{A} \times_1 \boldsymbol{P}_1 \times_2 \boldsymbol{P}_2 \times_3 \boldsymbol{S}\right\|_F, \left\|\mathcal{Y} - \mathcal{A} \times_3 \boldsymbol{P}_3\boldsymbol{S}\right\|_F, \left\{\left\|\mathcal{G}_n - \mathcal{A} \times_n \boldsymbol{D}_{I_n}\right\|_F\right\}_{n=1}^{2}\right\} > \epsilon$ **do**
  Update $\mathcal{A}$ via Eq. (10).
  for $n = 1 : 2$ **do**
    Update $\mathcal{G}_n$ via Eq. (14).
  end for
  Update $\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_x$ and $\mathcal{M}_y$ via Eq. (15).
  $\rho \leftarrow \nu\rho$.
end while
$\widehat{\mathcal{Z}} \leftarrow \mathcal{A} \times_3 \boldsymbol{S}$.

the sequence generated by Algorithm 1. If the sequences $(\{\mathcal{M}_{x,t}\}_{t\in\mathbb{N}})$ and $(\{\mathcal{M}_{y,t}\}_{t\in\mathbb{N}})$ are bounded, then the following statements hold:

1) $\{\mathcal{A}_t, \{\mathcal{G}_{n,t}\}_{n=1,2}\}_{t\in\mathbb{N}}$ is Cauchy sequence.
2) $\mathfrak{S}_t$ is bounded.
3) Any accumulation point of $\mathfrak{S}_t$, denoted as

$$\mathfrak{S}_* = \left\{\mathcal{A}_*, \{\mathcal{G}_{n,*}\}_{n=1,2}, \mathcal{M}_{x,*}, \mathcal{M}_{y,*}, \{\mathcal{M}_{n,*}\}_{n=1,2}\right\}$$

is a feasible point, i.e., it satisfies

$$\begin{cases} \mathcal{M}_{n,*} \in -\frac{1}{2}\partial \left\|\mathcal{G}_{n,*}\right\|_{3-n,\psi}, n = 1, 2 \\ \nabla L_1(\mathcal{A}_*) = 0 \\ \mathcal{X} = \mathcal{A}_* \times_1 \boldsymbol{P}_1 \times_2 \boldsymbol{P}_2 \times_3 \boldsymbol{S} \\ \mathcal{Y} = \mathcal{A}_* \times_3 \boldsymbol{P}_3\boldsymbol{S} \\ \mathcal{G}_{n,*} = \mathcal{A}_* \times_n \boldsymbol{D}_{I_n}, n = 1, 2. \end{cases}$$

Proof: Deferred to Section II in Supplemental Material. ■

Specifically, the boundedness of the Lagrangian multipliers is a commonly adopted assumption in convergence analysis, e.g., [56, Th. 5]. However, this condition may not hold when the penalty parameter is not properly increased, the constraint violations cannot be effectively reduced, or the optimization subproblems fail to produce stable updates. In general, establishing convergence guarantees for multivariable nonconvex optimization problems under fully characterized conditions remains a challenge. In our experiments, no convergence behavior violating this boundedness assumption was observed, which supports the practical validity of this condition.

IV. Experiments

In this section, we empirically validate the proposed method's effectiveness for the HSR task. Both nonblind and blind HSR are tested, among which the degradation matrices for blind HSR are estimated via [9]. The unsupervised state-of-the-art methods, including model-based Hysure [9], LTTR [19], LRTA [23], ASLA [26], GTNN [21], data-driven SURE [35], and ZSL [37], are involved for comparison. All model-based methods are implemented in MATLAB R2020b on Intel[1] Core[2] i5-1135G7 CPU at 2.40 GHz with 16-GB RAM, while the data-driven ones are implemented by Python 3.11.5 with Pytorch 1.10.2 on an NVIDIA RTX 3090 GPU with 24-GB RAM. The performance is measured in terms of peak signal-to-noise ratio (PSNR), relative dimensionless global error in synthesis (ERGAS), spectral angle mapper (SAM), and structural similarity metric (SSIM) [57]. The higher the PSNR and SSIM, the better the performance. The lower the ERGAS and SAM, the higher the performance. The ideal values are $+\infty$, 0, 0, and 1 for PSNR, ERGAS, SAM, and SSIM, respectively.

A. Datasets Description

Three hyperspectral datasets, i.e., URBAN,[3] Houston,[4], and Washington DC (WDC)[5] are employed. For URBAN, the original image contains 210 spectral bands and 307 $\times$ 307 spatial samples with a 2-m ground sampling distance. Following common preprocessing, the bands affected by water absorption and atmospheric interference were discarded, and a 256 $\times$ 256 $\times$ 162 subcube was cropped as the reference HSSI.

For Houston, the invalid zero-valued samples were first removed to obtain a 322 $\times$ 1903 $\times$ 144 cube. A 256 $\times$ 256 $\times$ 144 subscene was then cropped as the reference image. Since the original scene is visually dark, an elementwise gamma correction with exponent 0.7 was applied before generating the GT HSSI.

Since WDC was collected by the same sensor as URBAN, the same band-removal and cropping strategy was adopted, yielding a 256 $\times$ 256 $\times$ 162 reference HSSI.

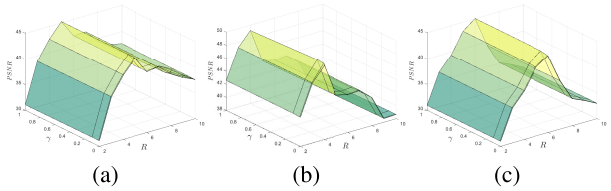
To synthesize the HSI, each reference HSSI was degraded using a 9 $\times$ 9 Gaussian blur kernel with standard deviation 3.3973, followed by 8$\times$ spatial decimation. The standard deviation is chosen, such that the kernel's frequency response at the Nyquist frequency $f_c = 1/(2 \times 8)$ is attenuated to e^{-1}. For URBAN and WDC, the MSI of size 256 $\times$ 256 $\times$ 6 was generated by averaging HSSI bands within six spectral ranges (450–520, 520–600, 630–690, 760–900, 1550–1750, and 2080–2350 nm) to match the spectral response of the USGS/NASA Landsat 7 satellite [50]. For Houston, the corresponding 256 $\times$ 256 $\times$ 4 MSI was generated via an IKONOS-like reflectance spectral response filter [25].

B. Experimental Results

1) Hyperparameter Sensitivity: The main hyperparameters include the dimensionality of the spectral subspace, i.e., R that captures the spectral low rankness of the latent HSSI and the surrogate parameter γ from $\psi(x) = (\log(\gamma x + 1))/(\log(\gamma + 1))$. To determine their values, we examine the PSNR sensitivity in

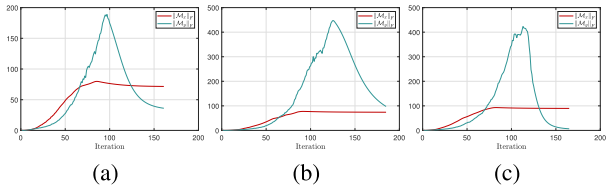
[1]Registered trademark.
[2]Trademarked.
[3]https://www.erdc.usace.army.mil/Media/Fact-Sheets/Fact-Sheet-Article-View/Article/610433/hypercube/#
[4]https://hyperspectral.ee.uh.edu/?page_id=459
[5]https://github.com/liangjiandeng/HyperPanCollection

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(a) (b) (c)

Fig. 2. Hyperparameter sensitivity reflected by the PSNR results for nonblind fusion on (a) URBAN, (b) Houston, and (c) WDC.



(a) (b) (c)

Fig. 3. Behavior of $\|\mathcal{M}_x\|_F$ and $\|\mathcal{M}_y\|_F$ throughout the iteration on (a) URBAN, (b) Houston, and (c) WDC.

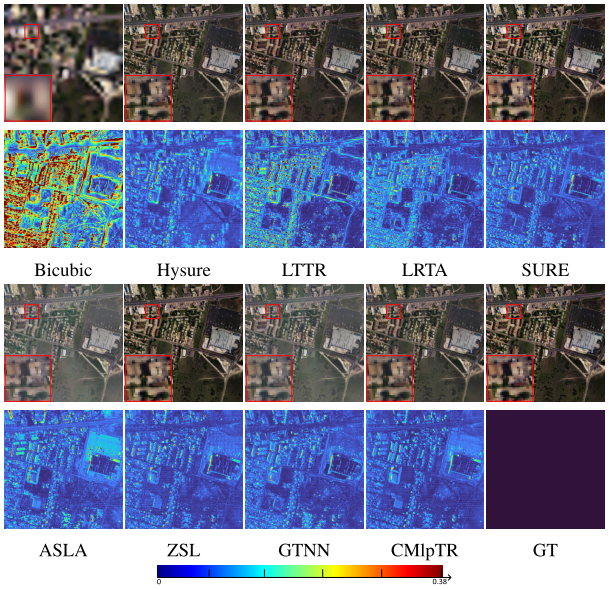


Fig. 4. Nonblind fusion results and error maps on the URBAN dataset. Pseudo-color is composed of bands 40, 30, and 10. Error maps are calculated by the pixelwise SAM.

nonblind fusion to both R and γ. The results are visualized in Fig. 2. It is clear to see that the proposed model is much more sensitive to R than to γ. From the distinct peak lines regarding R, it is easy to set R to five, four, and six for URBAN, Houston, and WDC, respectively. For γ, though the performance with respect to it remains relatively steady, there exists a black edge in each of the three surfaces at one common position where $\gamma = 0.1$. The corresponding PSNR value is higher than its neighbors if R is kept optimal. Thus, throughout these experiments, we set γ to 0.1.

2) Convergence Behavior: According to Theorem 1, the convergence of CMlpTR is guaranteed, provided that the Lagrangian multipliers $\mathcal{M}_x$ and $\mathcal{M}_y$ remain bounded. Hence, its convergence behavior can be assessed by monitoring $\|\mathcal{M}_x\|_F$ and $\|\mathcal{M}_y\|_F$. As illustrated in Fig. 3, both $\|\mathcal{M}_x\|_F$ and $\|\mathcal{M}_y\|_F$ are well bounded across all datasets, indicating

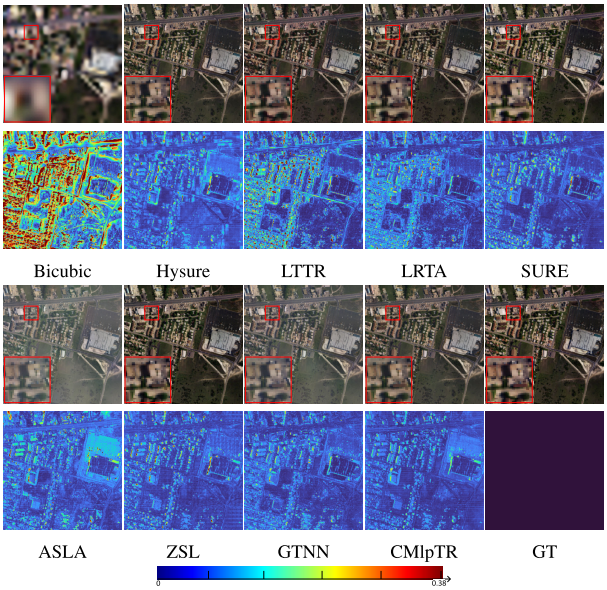


Fig. 5. Blind fusion results and error maps on the URBAN dataset. Pseudo-color is composed of bands 40, 30, and 10. Error maps are calculated by the pixelwise SAM.

TABLE I

NUMERICAL PERFORMANCE ON THE URBAN DATASET. THE BEST RESULTS ARE IN BOLDFACE

Methods	Setup							
	Non-blind				Blind			
	PSNR↑	ERGAS↓	SAM↓	SSIM↑	PSNR↑	ERGAS↓	SAM↓	SSIM↑
Bicubic	22.4646	5.8732	10.6064	0.4115	22.4646	5.8732	10.6064	0.4115
Hysure	42.9189	0.6890	2.9550	0.9920	42.7903	0.6778	2.9662	0.9920
LTTR	43.9124	0.8584	3.7677	0.9790	43.7321	0.8516	3.7948	0.9789
LRTA	44.5726	0.7171	2.8548	0.9873	44.3073	0.7187	2.8630	0.9873
SURE	42.9270	0.6169	2.4358	0.9915	42.7237	0.6326	2.5093	0.9914
ASLA	44.6256	0.6703	2.8596	0.9917	44.3815	0.6803	2.7664	0.9920
ZSL	43.1704	0.6226	2.4037	0.9927	42.9792	0.6431	2.5551	0.9921
GTNN	44.6832	0.5864	2.3858	0.9914	44.5186	0.5963	2.3980	0.9914
CMlpTR	**45.2351**	**0.5588**	**2.2375**	**0.9947**	**45.1130**	**0.5610**	**2.2425**	**0.9947**

that the boundedness condition is easily satisfied and thereby confirming the overall convergence of CMlpTR.

3) HSR Results on URBAN Dataset: The visual results are displayed in Figs. 4 and 5. In the nonblind results, all methods end up with fine spatial details, and only ASLA and GTNN exhibit slight color deviations and appear hazy. Besides, SURE, ZSL, GTNN, and the proposed CMlpTR generate the most darkest error maps, among which our CMlpTR is the best at suppressing higher error values. Similar conclusions can be drawn from the blind results in Fig. 5, except that all error maps have become brighter due to the estimation inaccuracies in the degradation matrices. The conclusions above are further strengthened by the numerical results in Table I, in which the proposed CMlpTR yielded the best values on all indexes, while SURE, ZSL, and GTNN are the most competitive having a look at the SAM metric.

4) HSR Results on Houston Dataset: The results on the Houston dataset are depicted in Figs. 6 and 7 and numerically reported in Table II. By the overall brightness of the error maps, it appears that ASLA, ZSL, and the proposed CMlpTR deliver the closest reconstruction. However, the ZSL suffers from block-like artifacts and ASLA is not as good as

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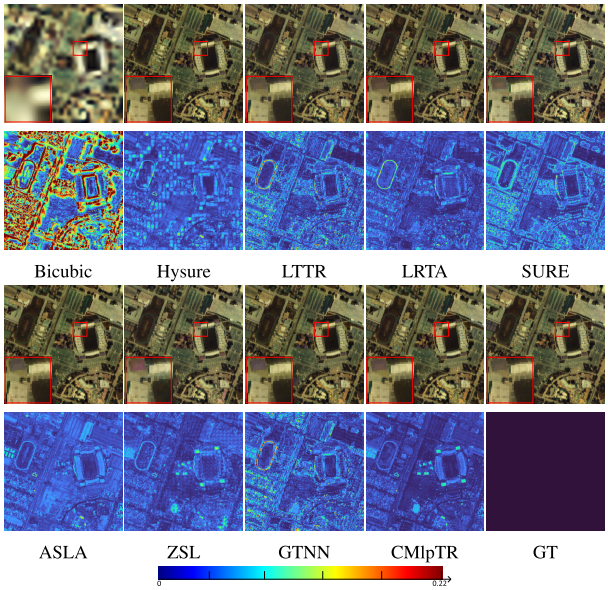


Fig. 6. Nonblind fusion results and error maps on the Houston dataset. Pseudo-color is composed of bands 45, 25, and 10. Error maps are calculated by the pixelwise SAM.

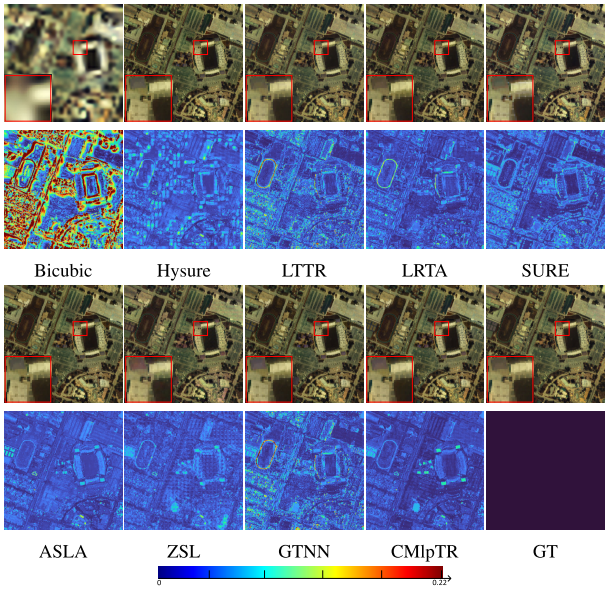


Fig. 7. Blind fusion results and error maps on the Houston dataset. Pseudo-color is composed of bands 45, 25, and 10. Error maps are calculated by the pixelwise SAM.

our CMlpTR in background suppression. Even though TV constraints are considered in ASLA, the multiconstraint paradigm leaves the problem of mutual interference. Besides, hinted by the numerical results in Table II, it is noticed that LRTA is better at reconstructing the bright regions in our CMlpTR's error map, leading to its competitiveness observing the SAM metric, even though LRTA shows random high-error pixels due to the neglect of TV priors.

5) HSR Results on WDC Dataset: The results on the WDC dataset are depicted in Figs. 8 and 9 and numerically reported in Table II. These visual results demonstrate that the WDC dataset presents greater textural complexity compared

TABLE II

NUMERICAL PERFORMANCE ON THE HOUSTON DATASET. THE BEST RESULTS ARE IN BOLDFACE

Methods	Setup							
	Non-blind				Blind			
	PSNR↑	ERGAS↓	SAM↓	SSIM↑	PSNR↑	ERGAS↓	SAM↓	SSIM↑
Bicubic	27.9764	2.8896	6.5494	0.6230	27.9764	2.8896	6.5494	0.6230
Hysure	48.2238	0.5736	1.4917	0.9927	46.3990	0.5866	1.5930	0.9922
LTTR	48.1071	0.5894	1.6414	0.9793	46.8301	0.5943	1.7098	0.9793
LRTA	46.7599	0.4785	1.1938	0.9922	46.5129	0.4790	1.3003	0.9919
SURE	44.3703	0.8480	1.6479	0.9769	44.8562	0.7868	1.4291	0.9829
ASLA	48.7790	0.8554	1.5664	0.9933	47.6438	0.7465	1.3068	0.9943
ZSL	46.1368	0.5463	1.1997	0.9911	45.5776	0.5667	1.3697	0.9912
GTNN	48.2029	0.5980	1.5338	0.9851	46.7844	0.6262	1.6013	0.9848
CMlpTR	**49.7612**	**0.4334**	**0.9367**	**0.9953**	**48.4184**	**0.4591**	**1.0784**	**0.9946**

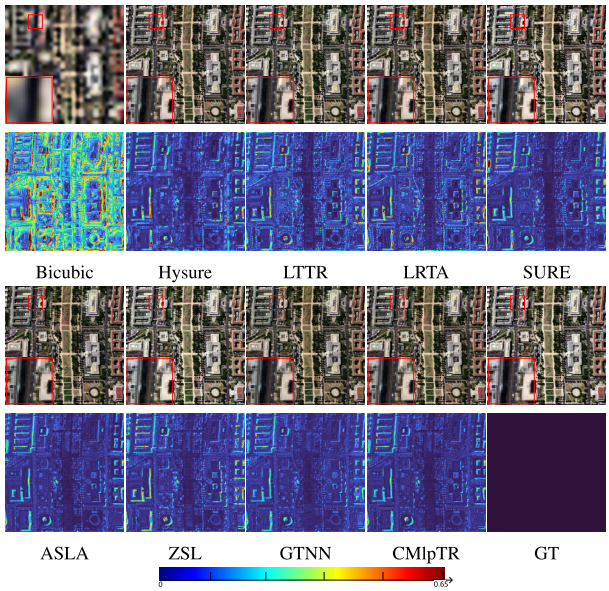


Fig. 8. Nonblind fusion results and error maps on the WDC dataset. Pseudo-color is composed of bands 40, 30, and 20. Error maps are calculated by the pixelwise SAM.

to the other two datasets, as evidenced by the pronounced high-frequency patterns in the error maps and the extended dynamic range of the colorbar. In Table III, the proposed CMlpTR performs the best, while ASLA turns out to be the most competitive achieving the best PSNR value in the non-blind experiment. The visual results agree with the numerical ones, where ASLA, GTNN, and CMlpTR obtain the darkest error maps.

C. Further Discussion

Table IV reports the runtime comparison of different methods under nonblind and blind settings. To be fair, only methods deployed onto CPU are considered. It is observed that the proposed method achieves a reasonable computational cost compared with the existing tensor-based approaches. Although some methods require less time due to simpler modeling strategies, the proposed method introduces more comprehensive tensor priors and adaptive optimization, leading to improved reconstruction performance. Moreover, the runtime variation under different degradation settings remains limited,

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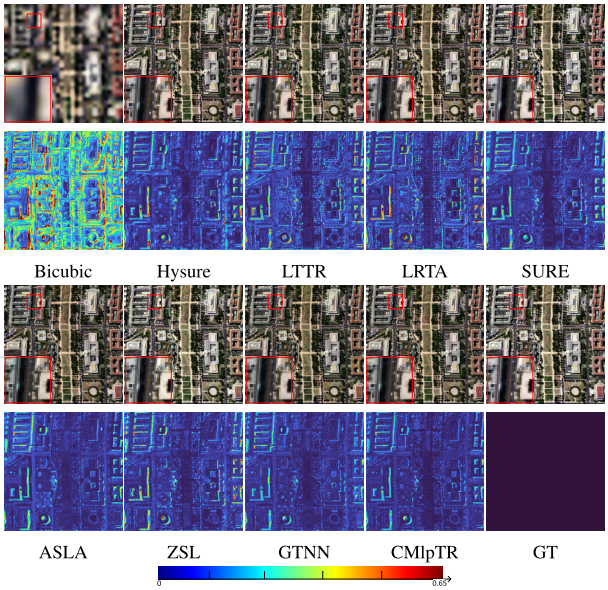


Fig. 9. Blind fusion results and error maps on the WDC dataset. Pseudo-color is composed of bands 40, 30, and 20. Error maps are calculated by the pixelwise SAM.

TABLE III

NUMERICAL PERFORMANCE ON THE WDC DATASET. THE BEST RESULTS ARE IN BOLDFACE

Methods	Setup							
	Non-blind				Blind			
	PSNR↑	ERGAS↓	SAM↓	SSIM↑	PSNR↑	ERGAS↓	SAM↓	SSIM↑
Bicubic	19.6920	7.1549	12.3534	0.2588	19.6920	7.1549	12.3534	0.2588
Hysure	43.6645	0.6142	3.2399	0.9936	43.4473	0.6048	3.2742	0.9935
LTTR	42.9732	0.7308	3.5122	0.9886	42.7777	0.7406	3.6109	0.9883
LRTA	43.8438	0.6604	3.6210	0.9890	43.6905	0.6624	3.6378	0.9890
SURE	42.2816	0.5884	2.9213	0.9931	42.0958	0.5961	2.8804	0.9928
ASLA	**45.1584**	0.5735	2.4119	0.9958	44.3206	0.5838	2.7515	0.9950
ZSL	42.3886	0.5939	2.9659	0.9935	42.9526	0.6144	3.1297	0.9932
GTNN	43.2923	0.5920	2.5136	0.9949	43.2412	0.5917	2.5163	0.9949
CMlpTR	45.0320	**0.4819**	**2.2969**	**0.9959**	**44.9592**	**0.4817**	**2.2971**	**0.9959**

TABLE IV

RUNTIME IN SECOND OF DIFFERENT METHODS

Methods	URBAN		Houston		WDC	
	Non-blind	Blind	Non-blind	Blind	Non-blind	Blind
Hysure	26.0602	21.6302	21.7921	22.6939	22.2027	23.7589
LTTR	90.5384	84.8955	44.8779	45.8387	68.8827	76.4060
LRTA	68.2543	73.0147	67.7208	66.3385	117.1862	119.3746
ASLA	54.3367	58.6080	48.2702	47.3190	51.6621	48.8079
GTNN	109.9113	99.8544	52.3368	52.9925	54.3946	55.2374
CMlpTR	67.9510	66.0787	62.6999	62.9548	84.4217	77.8856

demonstrating the efficiency and practicality of the proposed framework for HSR.

The difference matrices $\boldsymbol{D}_{l_n}$s are defined as aperiodic in the proposed model. To investigate the applicability of CMlpTR to periodic difference matrices, which are sometimes preferred due to their circulant structure, Fig. 10 presents the nonblind fusion results obtained with aperiodic (Row 1) and periodic (Row 2) difference matrices. It can be observed that CMlpTR achieves comparable reconstruction performance in both cases, demonstrating its robustness to different difference matrix configurations.

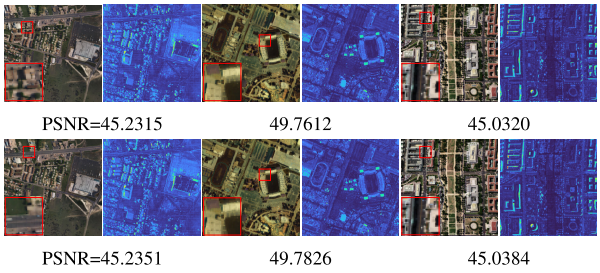


Fig. 10. Nonblind fusion results of CMlpTR using (row 1) aperiodic and (row 2) periodic difference matrices. The colorbars are consistent with those in Figs. 4, 6, and 8.

V. CONCLUSION

This article raised the attention on the difficulty in the co-representation of multilevel priors in the HSR task. Through the proposal of our NMP-t-CTV and combining it with the BTD, we achieved compact and tight representations of multilevel and multidimensional structural correlations inherent in HSSI while maintaining theoretical soundness. To efficiently solve the resulting high-dimensional optimization problem, an LADMM-inspired algorithm was customized, with convergence guarantees established under practical boundedness conditions. The compactness of the proposed model not only facilitated convergence analysis but also saved it from the problem of mutual interference between multiple constraints. Extensive experiments on both nonblind and blind HSR tasks demonstrated the practical effectiveness of the proposed approach, showing improved performance in preserving spectral-spatial details while keeping computational demands manageable.

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